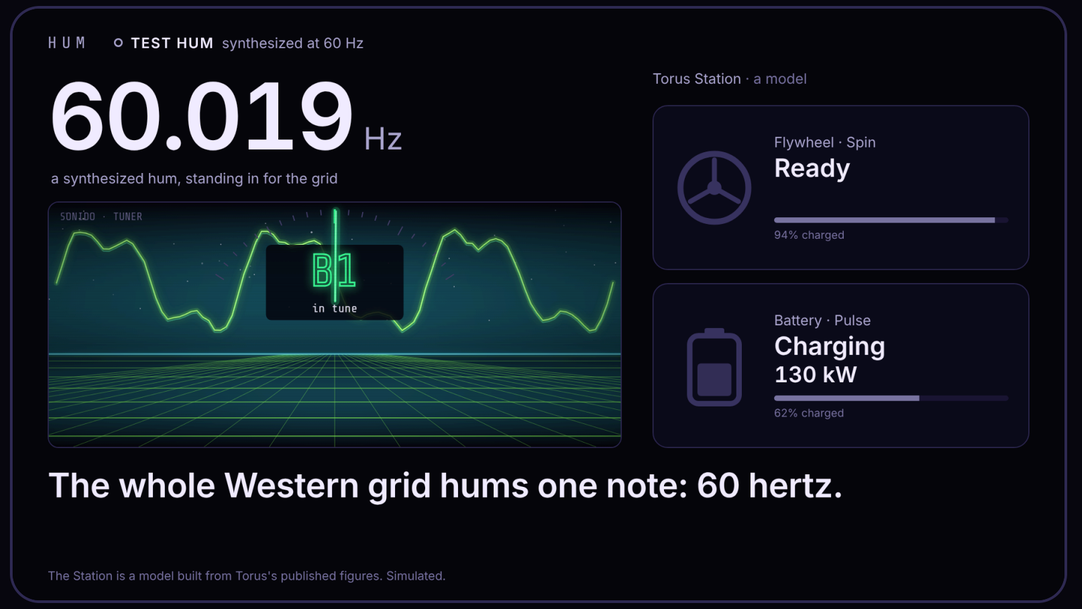
HUM: the power grid, heard through a guitar, and a Torus Station deciding what to do about it.

Every wire in a building hums at the grid's frequency. That one number is shared by the whole Western grid, from Alberta to Baja.

A guitar pickup hears it. My audio library, sonido, reads it to a thousandth of a hertz and shows it on its tuner.

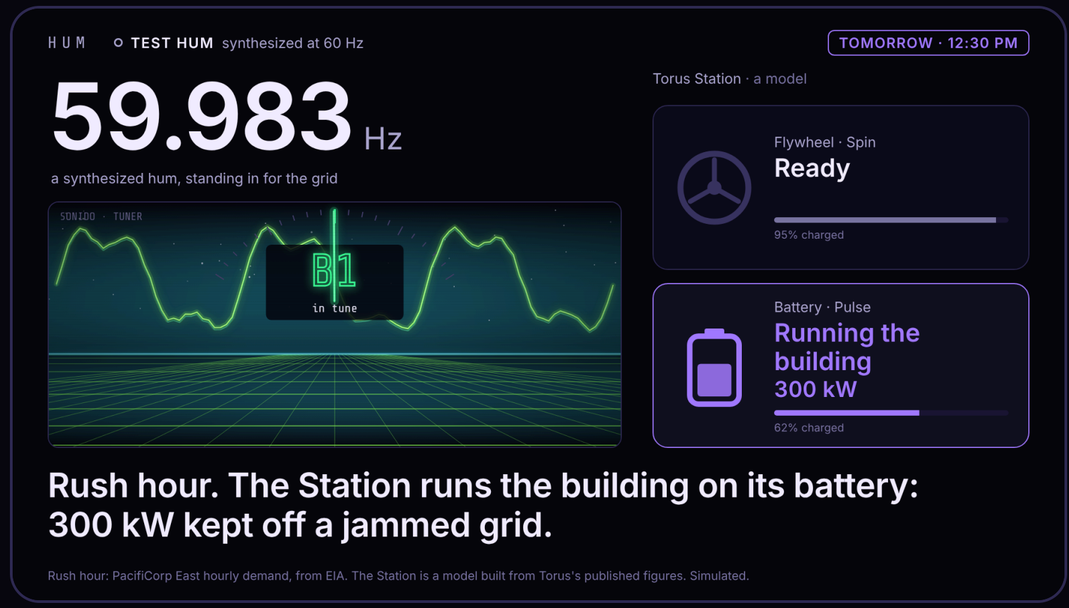


Rush hour you can see coming

The battery carries the building through the jam.

EIA's hourly data for PacifiCorp East shows when the grid gets jammed: this week, from noon to four.

At tomorrow's noon the Station runs the building on its battery, **300 kW kept off the grid**, and recharges when the power is clean.



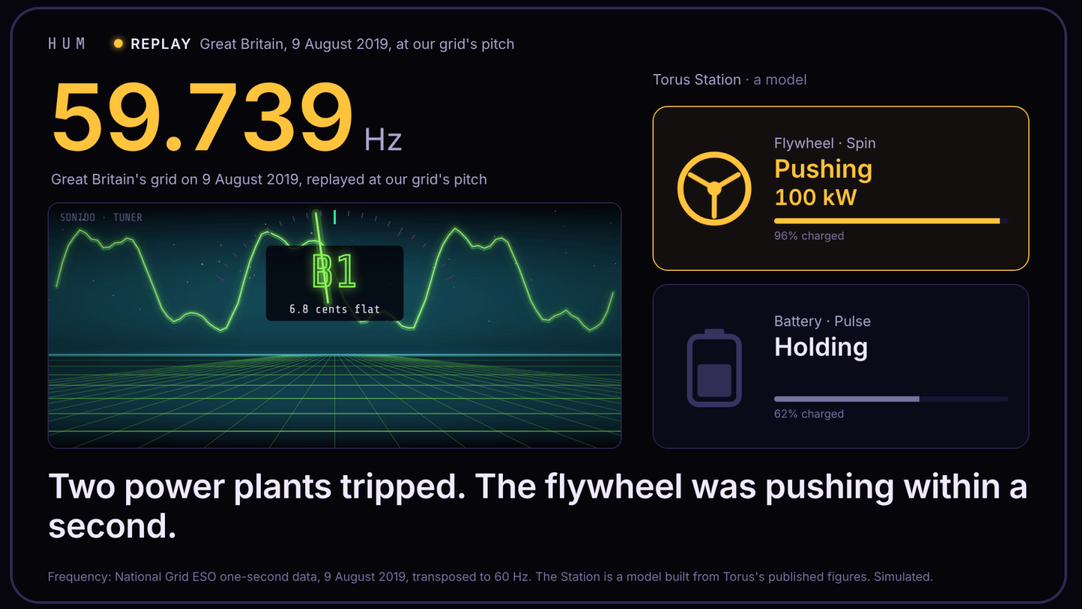
Data: EIA-930, PacifiCorp East. The Station is a model built from Torus's published figures.

A crash you can only hear

The flywheel answers within a second.

On 9 August 2019, two power plants in Britain tripped within a second and a million people lost power. Its real frequency record, replayed at our grid's pitch, pulls the number down.

The flywheel pushes at full power before an hourly feed could notice. **When the grid goes dark, the building stays on.**



Data: National Grid ESO one-second frequency, 9 August 2019, transposed to 60 Hz.

For your engineers

The software making these decisions can't reach the internet.

It is written in Mentl, a language I built in Utah. Its compiler proves that promise through every function the decision calls. Add one line that reads a price online, and it refuses to build the program.

✕ Refused

One line broke the promise.

```
let price = price_m("https://api.eia.gov/…")
```

This program promised it can't reach the internet while it decides. Add one line that reads a price online, and the compiler won't build it. Its error names the path the price took.

Replay the demo:
`hum.ampactor.dev`
sonofagoose@pm.me

Thank you, Torus.